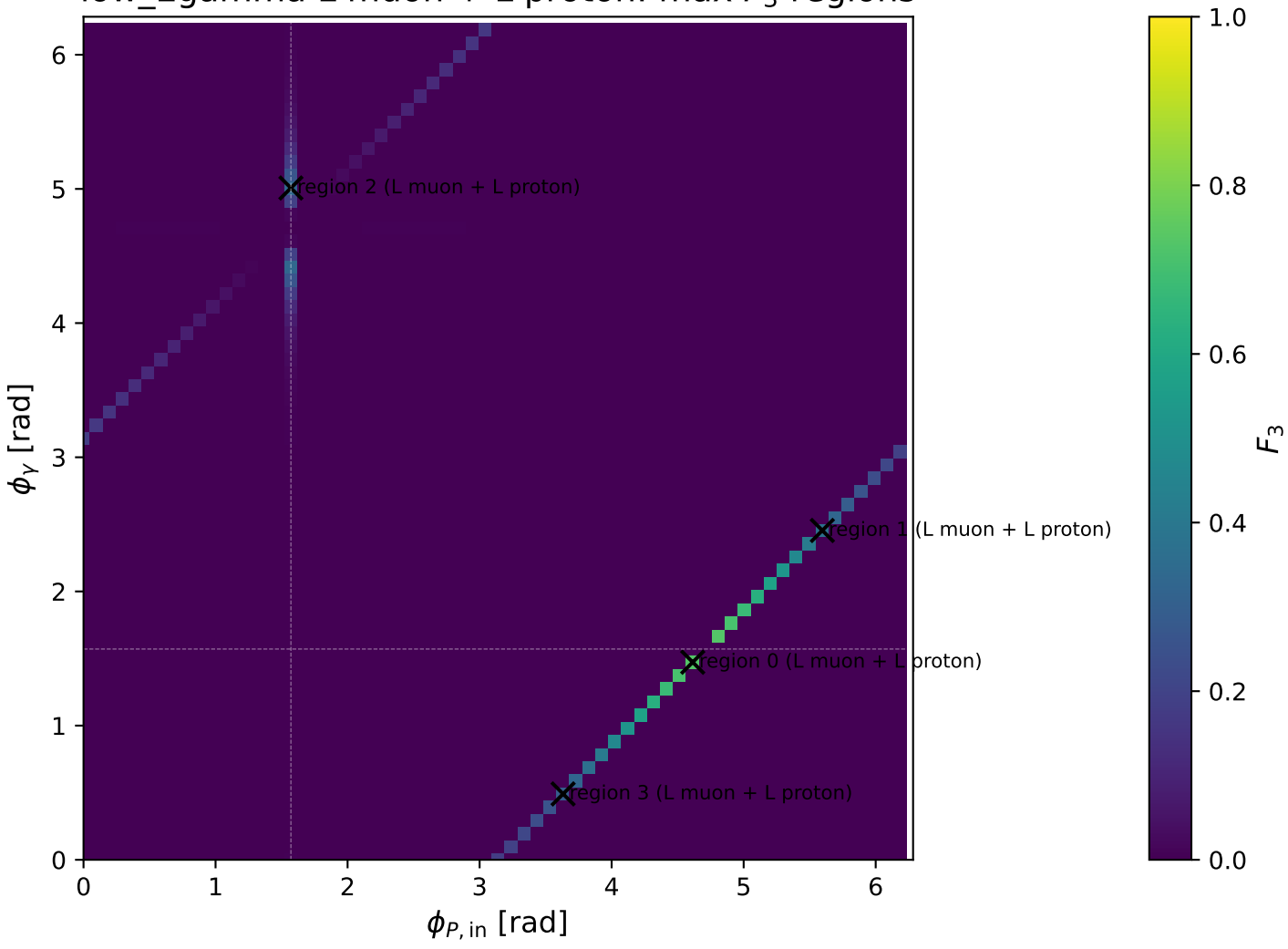
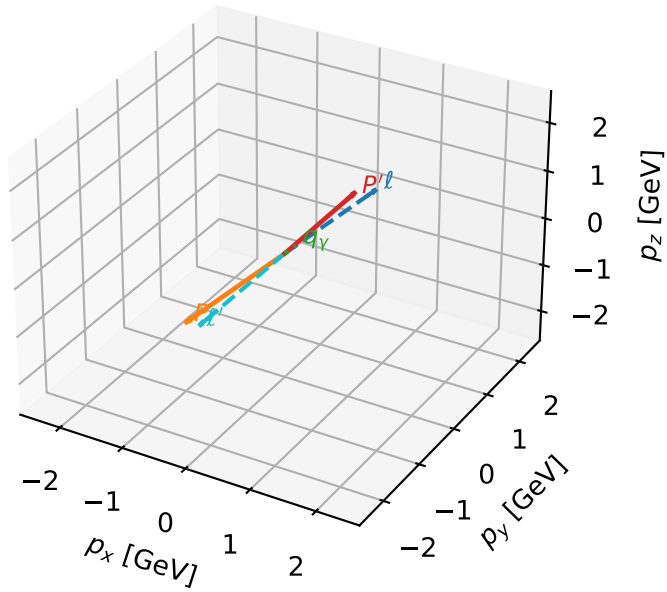


low_Egamma L muon + L proton: max F_3 regions



L muon + L proton characteristic max F_3 configuration

3D momenta

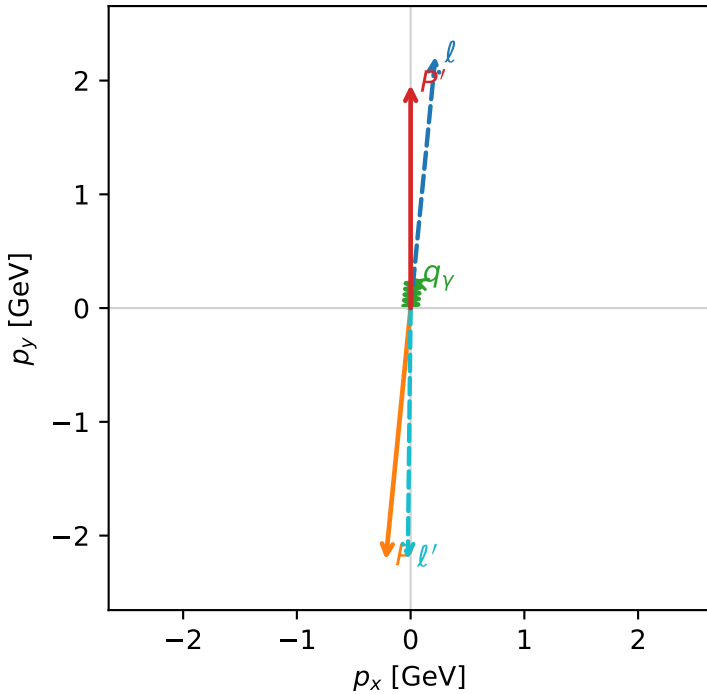


f3_low_egamma_ll_region_0 $F_3=0.730522$
 region: low_Egamma_LL_region_0
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_low_Egamma
 incoming state: $|L_\mu\rangle \otimes |L_p\rangle$

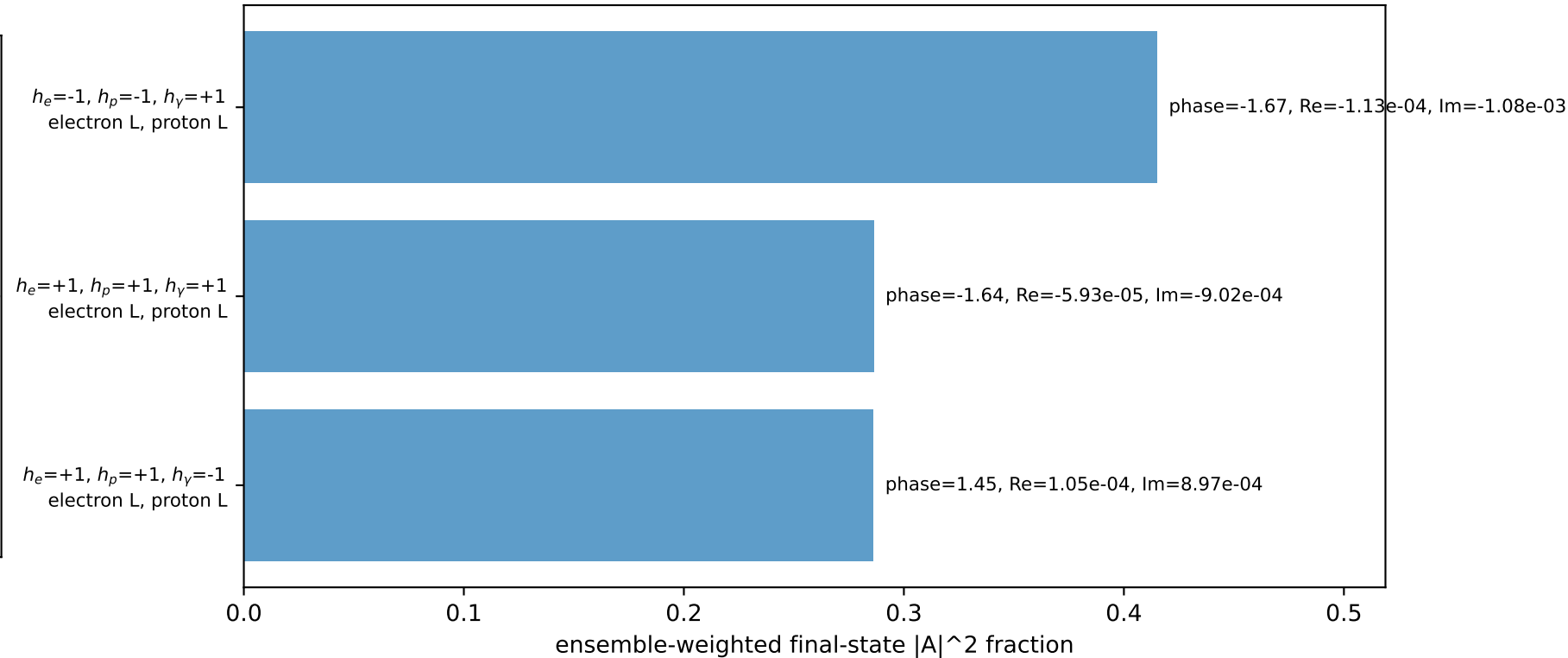
$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22311$, $|\vec{P}'|=1.96219$
 $\theta_{in}=1.57079$, $\phi_l=1.47262$, $\phi_p=4.61421$
 $E_\gamma=0.25$, $\phi_\gamma=1.47262$
 $Q^2=19.6251$, $x_B=0.994373$, $t=-17.4181$, $W^2=0.990905$, $y=0.956911$
 $\theta(l', \gamma)=175.01$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.2256$ $p_x= 0.2179$ $p_y= 2.2124$ $p_z= -0.0000$
 P $E= 2.4129$ $p_x= -0.2179$ $p_y= -2.2124$ $p_z= 0.0000$
 l' $E= 2.2136$ $p_x= -0.0245$ $p_y= -2.2110$ $p_z= 0.0000$
 P' $E= 2.1749$ $p_x= 0.0000$ $p_y= 1.9622$ $p_z= 0.0000$
 q_γ $E= 0.2500$ $p_x= 0.0245$ $p_y= 0.2488$ $p_z= 0.0000$

Transverse plane

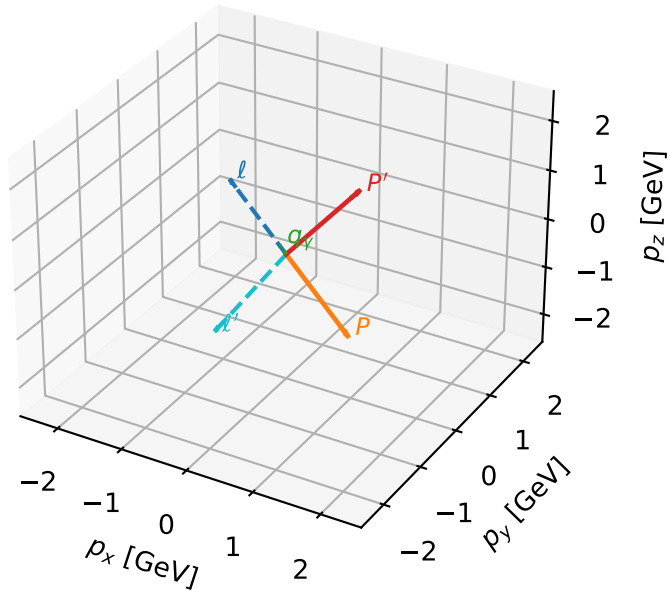


Leading initial-ensemble/final-state components (N=3, retained=98.8%)



L muon + L proton characteristic max F_3 configuration

3D momenta

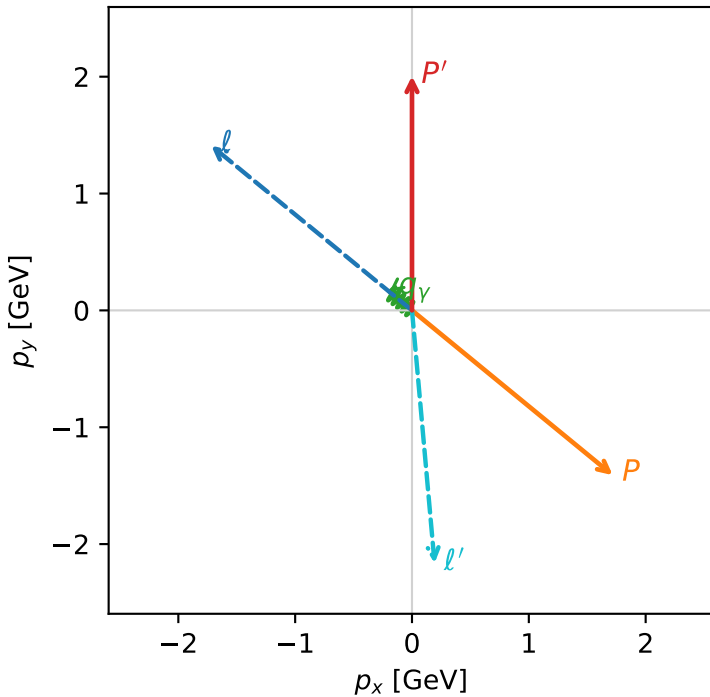


f3_low_egamma_ll_region_1 $F_3=0.367049$
 region: low_Egamma_LL_region_1
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_low_Egamma
 incoming state: $|L_\mu\rangle \otimes |L_p\rangle$

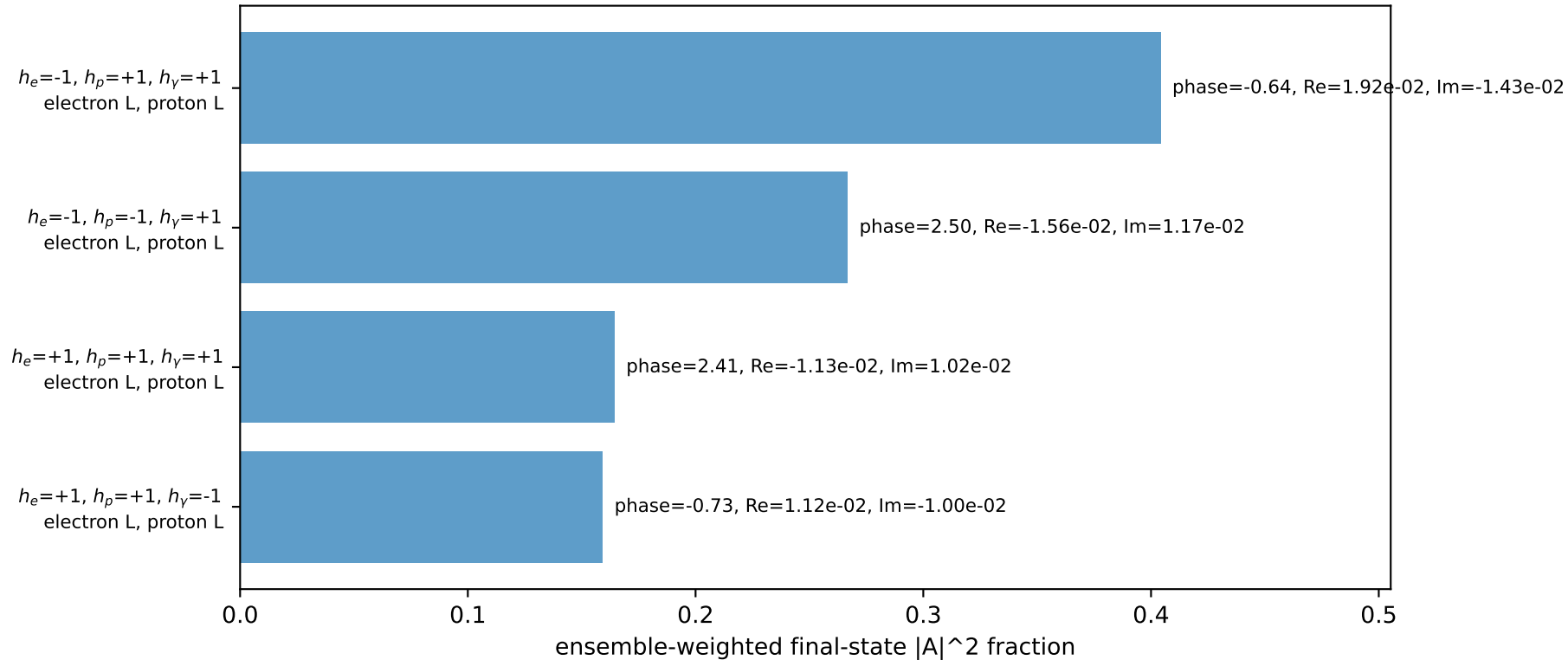
$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22311$, $|\vec{P}'|=2.00509$
 $\theta_{in}=1.57079$, $\phi_l=2.45437$, $\phi_P=5.59596$
 $E_\gamma=0.25$, $\phi_\gamma=2.45437$
 $Q^2=16.4258$, $x_B=0.972135$, $t=-14.5786$, $W^2=1.35066$, $y=0.819235$
 $\theta(l', \gamma)=134.479$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.2256$ $p_x= -1.7185$ $p_y= 1.4103$ $p_z= -0.0000$
 P $E= 2.4129$ $p_x= 1.7185$ $p_y= -1.4103$ $p_z= 0.0000$
 l' $E= 2.1749$ $p_x= 0.1933$ $p_y= -2.1637$ $p_z= 0.0000$
 P' $E= 2.2136$ $p_x= 0.0000$ $p_y= 2.0051$ $p_z= 0.0000$
 q_γ $E= 0.2500$ $p_x= -0.1933$ $p_y= 0.1586$ $p_z= 0.0000$

Transverse plane

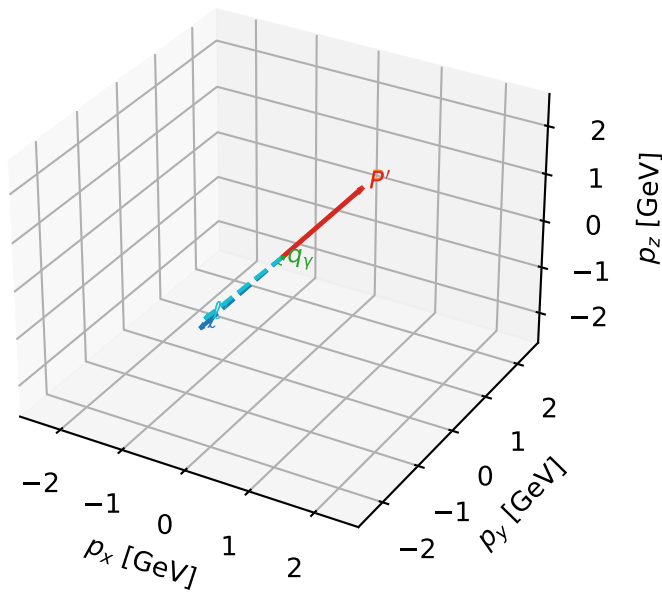


Leading initial-ensemble/final-state components (N=4, retained=99.5%)



L muon + L proton characteristic max F_3 configuration

3D momenta

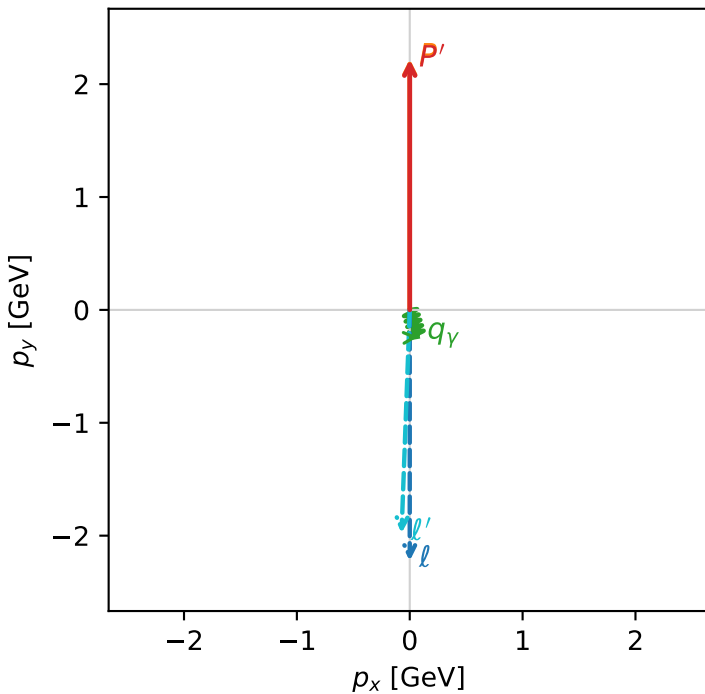


f3_low_egamma_ll_region_2 F_3=0.330468
region: low_Egamma_LL_region_2
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_low_Egamma
incoming state: $|L_\mu\rangle \otimes |L_p\rangle$

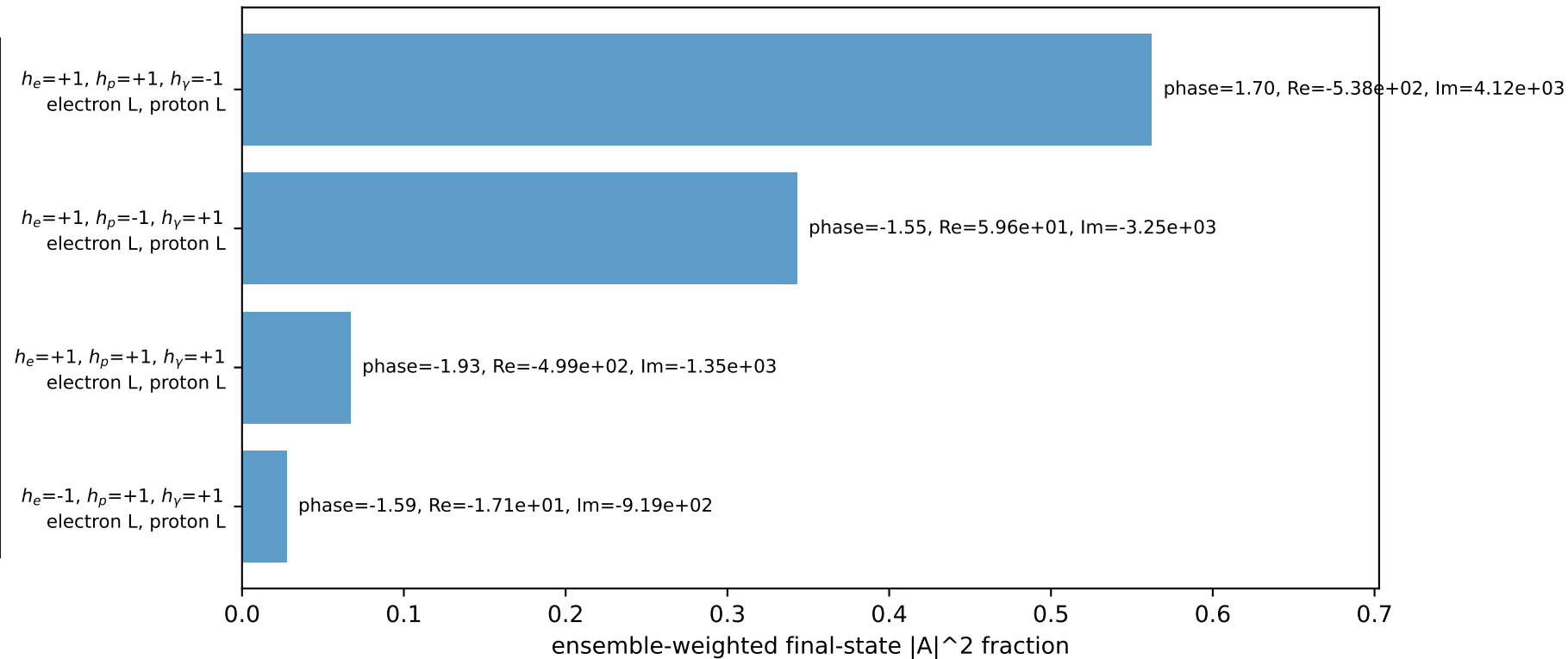
$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22311$, $|\vec{P}'|=2.21665$
 $\theta_{in}=1.57079$, $\phi_l=4.71239$, $\phi_p=1.5708$
 $E_\gamma=0.25$, $\phi_\gamma=5.00691$
 $Q^2=0.0060701$, $x_B=0.00267388$, $t=-6.31712e-06$, $W^2=3.14392$, $y=0.110069$
 $\theta(l', \gamma)=18.9768$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.2256$ $p_x= -0.0000$ $p_y= -2.2231$ $p_z= -0.0000$
 P $E= 2.4129$ $p_x= 0.0000$ $p_y= 2.2231$ $p_z= 0.0000$
 l' $E= 1.9816$ $p_x= -0.0726$ $p_y= -1.9774$ $p_z= 0.0000$
 P' $E= 2.4069$ $p_x= 0.0000$ $p_y= 2.2167$ $p_z= 0.0000$
 q_γ $E= 0.2500$ $p_x= 0.0726$ $p_y= -0.2392$ $p_z= 0.0000$

Transverse plane

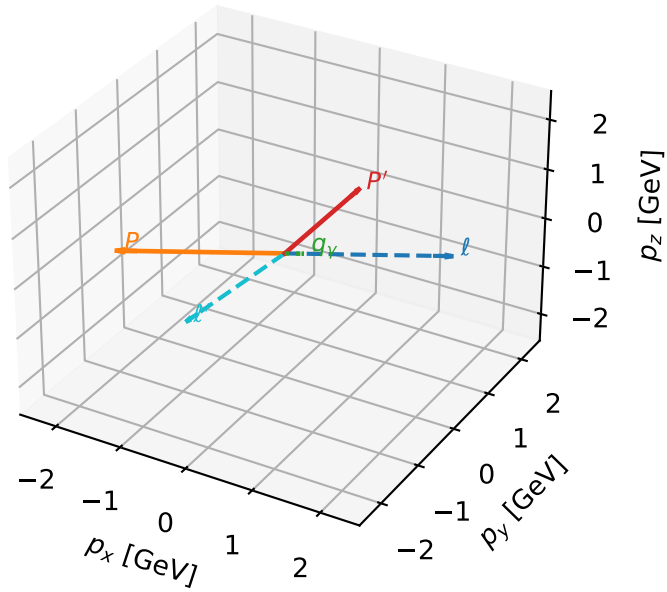


Leading initial-ensemble/final-state components (N=4, retained=100.0%)



L muon + L proton characteristic max F_3 configuration

3D momenta

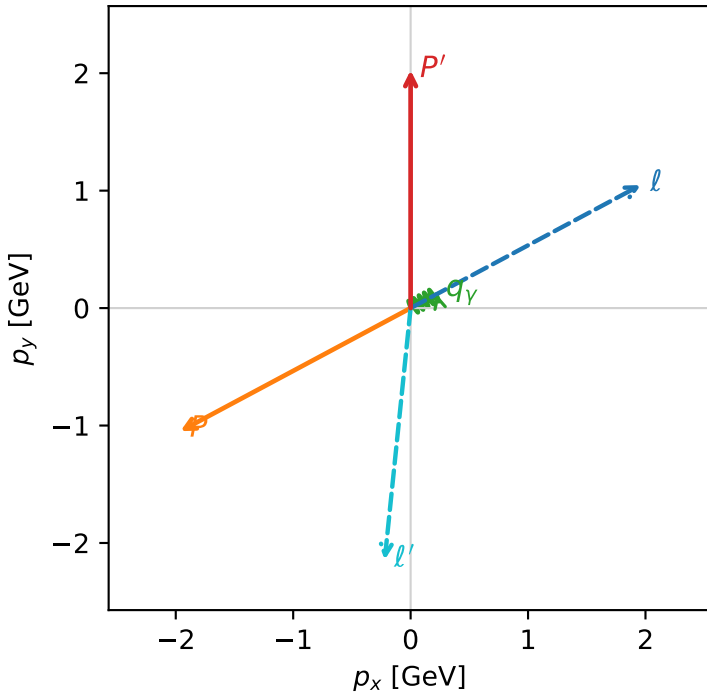


f3_low_egamma_ll_region_3 $F_3=0.290689$
 region: low_Egamma_LL_region_3
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_low_Egamma
 incoming state: $|L_\mu\rangle \otimes |L_p\rangle$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22311$, $|\vec{P}'|=2.02504$
 $\theta_{in}=1.57079$, $\phi_l=0.490874$, $\phi_p=3.63247$
 $E_\gamma=0.25$, $\phi_\gamma=0.490874$
 $Q^2=14.934$, $x_B=0.958994$, $t=-13.2545$, $W^2=1.51841$, $y=0.755038$
 $\theta(l', \gamma)=123.999$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.2256$ $p_x= 1.9606$ $p_y= 1.0480$ $p_z= -0.0000$
 P $E= 2.4129$ $p_x= -1.9606$ $p_y= -1.0480$ $p_z= 0.0000$
 l' $E= 2.1568$ $p_x= -0.2205$ $p_y= -2.1429$ $p_z= 0.0000$
 P' $E= 2.2317$ $p_x= 0.0000$ $p_y= 2.0250$ $p_z= 0.0000$
 q_γ $E= 0.2500$ $p_x= 0.2205$ $p_y= 0.1178$ $p_z= 0.0000$

Transverse plane



Leading initial-ensemble/final-state components (N=4, retained=99.2%)

